

$$1. v_1 = 1 \quad v_2 = x \quad v_3 = x^2 \Rightarrow (w_1, w_2, w_3)$$

a)

$$I. \quad w_1 = v_1 = 1 \quad \|w_1\| = \sqrt{\int_1^5 dx} = \sqrt{4} = 2 \quad \underline{\underline{w_1 = \frac{1}{2}}}$$

$$II. \quad w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 \quad \langle x, 1 \rangle = \int_1^5 x dx = \dots = 12$$

orthogonalize

$$\langle 1, 1 \rangle = 4$$

normalization:

$$w_{ij} = \frac{w_i}{\|w_i\|}$$

$$w_2 = x - 3$$

$$\|w_2\| = \sqrt{\int_1^5 (x-3)^2 dx} = \dots = \sqrt{\frac{16}{3}}$$

$$\underline{\underline{w_2 = \frac{x-3}{\sqrt{16/3}}}}$$

$$III. \quad w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

$$v_3 = x^2, w_1 = 1, w_2 = x - 3$$

$$\langle v_3, w_1 \rangle = \langle x^2, 1 \rangle = \int_1^5 x^2 dx = \dots = \frac{124}{3}$$

$$\langle w_1, w_1 \rangle = \langle 1, 1 \rangle = 4$$

$$\langle v_3, w_2 \rangle: \langle x^2, x-3 \rangle = \int_1^5 x^2(x-3) dx = \int_1^5 (x^3 - 3x^2) dx = \dots = 32$$

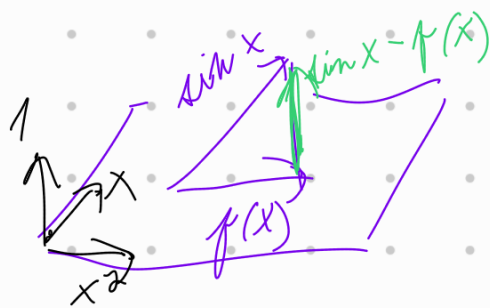
$$\langle w_2, w_2 \rangle = \frac{16}{3}$$

$$w_3 = x^2 - \frac{\frac{124}{3}}{4} \cdot 1 - \frac{32}{16/3} \cdot (x-3) = x^2 - 6x - \frac{23}{3}$$

$$\|w_3\| = \sqrt{\int_1^5 (x^2 - 6x - \frac{23}{3})^2 dx} = \dots = 2.39$$

$$\underline{\underline{w_3 = \frac{x^2 - 6x - \frac{23}{3}}{2.39}}}$$

b)



potni:

$$\langle \sin x - f(x), 1 \rangle = 0$$

$$\langle \sin x - f(x), x \rangle = 0$$

$$\langle \sin x - f(x), x^2 \rangle = 0$$

$$f(x) = ax^2 + bx + c$$

$$\int_1^5 (\sin x - (ax^2 + bx + c)) \cdot 1 dx = 0$$

$$\int_1^5 (\sin x - (ax^2 + bx + c)) \cdot x dx = 0$$

$$\int_1^5 (\sin x - (ax^2 + bx + c)) \cdot x^2 dx = 0$$

skupno:

$$\int_1^5 \sin x dx - a \int_1^5 x^2 dx - b \int_1^5 x dx - c \int_1^5 1 dx = 0$$

$$\int_1^5 x \sin x dx - a \int_1^5 x^3 dx - b \int_1^5 x^2 dx - c \int_1^5 x dx = 0$$

$$\int_1^5 x^2 \sin x dx - a \int_1^5 x^4 dx - b \int_1^5 x^3 dx - c \int_1^5 x^2 dx = 0$$

...

$$0.26 - 41.33a - 12b - 4c = 0$$

$$-2.68 - 156a - 41.33b - 12c = 0$$

$$-14.34 - 624.8a - 156b - 41.33c = 0$$

$$\underline{f(x) \approx -0.053x^2 - 0.33x + 1.60}$$

c) norma dyaly:

$$\| \sin x - f(x) \| = \sqrt{\int_1^5 (\sin x - f(x))^2 dx} = \dots \approx 0.32$$